

Microprocessor, microcontroller and embedded system

Course code: CSE-3523

Segment:01

Previous Question Solution:

01.Differentiate between CPU & Microprocessor. (2 marks, sp-23)

Answer:

Difference between CPU and Microprocessor:

CPU	Microprocessor
Central Processing Unit – the brain of the computer that performs calculations and control tasks.	A CPU built on a single Integrated Circuit (IC) chip.
May consist of multiple chips (ALU, CU, registers, etc.	All CPU components are packed into one chip.

Note: Every microprocessor is a CPU, but not every CPU is a microprocessor.

02.In the microprocessor, what is the function of IP and ALU? (2 marks, sp-22, Aut-23)

Answer:

IP (Instruction Pointer): It is a register in the microprocessor that always stores the memory address of the next instruction. After one instruction is executed, IP updates to point to the following instruction.

ALU (Arithmetic Logic Unit): It is the part of the microprocessor responsible for carrying out all arithmetic operations (like addition, subtraction, multiplication, division) and logical operations (like AND, OR, NOT, comparisons).

Note: IP → next instruction track core, ALU → calculation & logic core

03.What are the essential components of a microprocessor? what makes the microprocessor different from the microcontroller? (3 marks, **Aut-24**)

Answer:

Microprocessor	Microcontroller
Microprocessor is the heart of Computer system.	Microcontroller is the heart of an embedded system.
It is only a processor, so memory and I/O components need to be connected externally.	Microcontroller has a processor along with internal memory and I/O components.
Memory and I/O have to be connected externally, so the circuit becomes large.	Memory and I/O are already present, and the internal circuit is small.
You can't use it in compact systems.	You can use it in compact systems.
Cost of the entire system is high.	Cost of the entire system is low.

Note: Microprocessor = brain only

Microcontroller = brain + memory + peripherals (সহায়ক যন্ত্রাংশ) in one chip

04. What are the basic components of a microprocessor? what makes the microprocessor different from the CPU? which has more efficiency and how? (3 marks, sp-24)

Answer:

The basic components of a microprocessor are the Arithmetic Logic Unit (ALU) which performs all calculations and logic operations, registers which act as small storage for data and instructions, the control unit which directs the flow of data and operations, and the clock unit which provides timing signals. The difference between a CPU and a microprocessor is that the CPU contains only the ALU and the control unit, while the microprocessor includes the CPU along with extra circuits such as registers, cache, and the bus interface on a single chip. A microprocessor is considered more efficient because everything is integrated on one chip, resulting in faster data transfer, smaller size, and lower cost.

Note: Microprocessor has more efficiency because everything is integrated on one chip → reduces data transfer time & improves speed.

05. suppose you need to run a high quality and speedy performance video on computers. what type of bus you will use to support this operation on the computer? Describe the bus. (3 marks, Aut-23)

Answer:

To run a high-quality and speedy video on a computer, we use a **high-speed graphics bus**, such as the **Accelerated Graphics Port (AGP)** or the modern **PCI Express (PCIe) bus**.

A bus is a communication pathway that transfers data between the microprocessor, memory, and input/output devices. For video, a large amount of data must move very quickly. The **AGP bus** was designed especially for high-performance video; it connects directly to the CPU and uses the computer's main memory to store 3D images, which allows smooth video playback. In

modern systems, **PCIe** is used, which provides very high bandwidth, point-to-point connections, and fast transfer rates.

This kind of bus supports fast data flow between the graphics card and CPU, making high-resolution and high-frame-rate video possible without delay.

06. You are sitting at a specific place inside a bus. The ticket checker asks you to show your ticket. Do you see any similarity between address, control and data bus with the above-mentioned scenario. If yes, explain with logic (3 marks, sp-23)

Answer:

When I sit in my seat on a bus and the ticket checker comes to check my ticket, it is very similar to how a computer's buses work. My seat number is like the address bus because it tells the checker exactly where to find me. When the checker asks to see my ticket, it acts like the control bus, giving a signal for an action to be performed. The ticket I show is like the data bus because it carries the actual information from me to the checker. Just like this, in a computer, the address, control, and data buses work together to make sure the correct information reaches the right place at the right time, efficiently and without confusion.